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UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER TWO EXAMINATIONS, 2024/2025

TIME: 2 HOURS

COURSE CODE: TEUP 206

COURSE TITLE: TEACHING AND ASSESSING MATHEMATICS FOR UPPER PRIMARY

(INTRODUCTORY)

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and three from Section B.

SECTION A

[25 marks]

Answer all the questions in this section.

1. Which of the following best explains why the "friendly jump" strategy is effective for adding 49 + 36?
   A. It uses number bonds to 10
   B. It changes one addend to a multiple of 10 before adding
   C. It avoids carrying over digits
   D. It always produces the largest sum possible

2. In base-ten, the digit 7 in 7,845 represents:
   A. Seven ones
   B. Seven tens
   C. Seven hundreds
   D. Seven thousands

3. Which strategy is most efficient for solving 805 – 398 mentally?
   A. Compensation
   B. Counting all
   C. Take away
   D. Making doubles

4. Which set shows equivalent fractions?
   A. 3/4 and 6/9
   B. 5/8 and 10/16
   C. 4/5 and 12/20
   D. 2/3 and 5/9

5. The NACCA standards-based curriculum is different from the CRDD objective-based curriculum mainly because it:
   A. Focuses on teacher-led instruction
   B. Places more emphasis on competencies
   C. Requires less assessment
   D. Avoids real-life applications

6. Which manipulative is best for teaching part-whole relationships in fractions?
   A. Geoboard
   B. Fraction strips
   C. Abacus
   D. Protractor

7. If 3/5 of a class are girls and there are 18 girls, how many pupils are in the class?
   A. 24
   B. 27
   C. 30
   D. 36

8. Which is an example of AfL (Assessment for Learning)?
   A. Final exam in mathematics
   B. Teacher giving feedback during a lesson
   C. Standardized national test
   D. End-of-term project grading

9. When teaching subtraction, "counting up" works best when:
   A. The subtrahend is close to the minuend
   B. The numbers are far apart
   C. There is no regrouping
   D. The minuend is less than the subtrahend

10. Which model helps pupils understand that 0.5 is equivalent to 1/2?
   A. Hundreds chart
   B. Number line
   C. Place value chart
   D. Multiplication table

11. Which approach is most appropriate for addressing socio-cultural bias in mathematics?
    A. Avoiding local contexts
    B. Using only English in word problems
    C. Including examples from pupils' communities
    D. Focusing only on textbook exercises

12. Which operation does the "compensation" strategy adjust before computing?
    A. Only the minuend
    B. Only the subtrahend
    C. Either minuend or subtrahend
    D. Both minuend and subtrahend simultaneously

13. If 25% of 48 pupils are absent, how many are present?
    A. 12
    B. 24
    C. 36
    D. 40

14. The best reason for using manipulatives with SEN pupils is that they:
    A. Reduce lesson time
    B. Provide abstract representations
    C. Give concrete, hands-on experience
    D. Eliminate need for explanation

15. Which of these is NOT a mental addition strategy taught in upper primary?
    A. Friendly jumps
    B. Making doubles
    C. Constant difference
    D. Counting on

16. Which of these is the least effective assessment for detecting pupils' misconceptions in place value?
    A. One-on-one interview
    B. Multiple choice test
    C. Group project on number systems
    D. Asking pupils to represent numbers with base-ten blocks

17. A rectangle has length 8 cm and width 6 cm. If both dimensions are doubled, the area becomes:
    A. 48 cm^2
    B. 96 cm^2
    C. 192 cm^2
    D. 384 cm^2

18. Which of the following strategies would best help pupils solve 999 – 478 quickly?
    A. Making doubles
    B. Constant difference
    C. Compensation
    D. Friendly jumps

19. If 4/7 of the pupils like football and the rest like volleyball, what fraction like volleyball?
    A. 3/7
    B. 4/7
    C. 7/4
    D. 7/3

20. Which ICT tool is most appropriate for teaching multiplication facts interactively?
    A. Spreadsheet
    B. Drill-and-practice software
    C. Video streaming site
    D. Email platform

21. When adding 37 + 49, which method is an example of "friendly jumps"?
    A. 37 + 50 – 1
    B. 37 + 40 + 9
    C. 37 + 10 + 39
    D. 37 + 13 + 36

22. The key advantage of peer assessment is that it:
    A. Reduces teacher workload only
    B. Encourages pupils to reflect on their own work
    C. Ensures higher grades
    D. Removes the need for tests

23. In the constant difference method for subtraction, the same number is:
    A. Added to both numbers
    B. Subtracted from both numbers
    C. Added to the minuend and subtracted from the subtrahend
    D. Subtracted from the minuend and added to the subtrahend

24. The number 602 rounded to the nearest hundred is:
    A. 600
    B. 610
    C. 620
    D. 700

25. Which of the following is the MOST suitable real-life context for teaching skip counting by 25s?
    A. Counting apples in a basket
    B. Counting Ghana cedi coins of 25 pesewas
    C. Measuring time in hours
    D. Counting in dozens

SECTION B

[75 marks]

Answer three questions from this section.

1a. Explain, with examples, **THREE** (3) ways the NACCA standards-based curriculum for P4–P6 differs in focus from the CRDD objective-based curriculum. **[9 marks]**

1b. Design a short classroom activity (with steps) to help pupils mentally add 398 and 425. **[8 marks]**

1c. Critically evaluate one potential misconception pupils may have about zero in place value, and propose a strategy to address it. **[8 marks]**

2a. Solve: A class of 36 pupils has 2/3 wearing school jerseys and the rest in plain shirts. If 5/8 of those in jerseys are girls, how many boys are wearing jerseys? Show full workings. **[10 marks]**

2b. Justify the use of ICT tools for administering mathematics assessments in upper primary, citing at least **THREE** (3) specific advantages. **[15 marks]**

3a. Differentiate between AfL, AoL, and AaL. Give one example each. **[9 marks]**

3b. Design a 4-question mini-test on subtraction within 99 that meets the five criteria of a good test (clarity, validity, practicality, efficiency, fairness). State how each criterion is met. **[12 marks]**

3c. Using the "constant difference" method, solve 607 – 498. **[4 marks]**

4a. Identify two socio-cultural beliefs that could hinder girls' participation in mathematics learning in upper primary. **[8 marks]**

4b. A baker uses 3/4 of a sack of flour to bake bread in the morning and 2/5 of a sack in the afternoon. How much flour is used in total? Express your answer as a simplified fraction. **[8 marks]**

4c. A pupil claims that 7,004 rounded to the nearest hundred is 7,000. Is the pupil correct? Provide a detailed explanation using place value concepts. **[9 marks]**

5a. Solve: (345 + 256) − (427 − 189) using at least two mental strategies taught in P4–P6, and explain each.

[8 marks]

5b. Design a real-world context problem that requires pupils to use both addition and subtraction within 999. **[7 marks]**

5c. Suggest a peer-assessment rubric for evaluating solutions to your problem, focusing on correctness, method, and explanation. **[10 marks]**